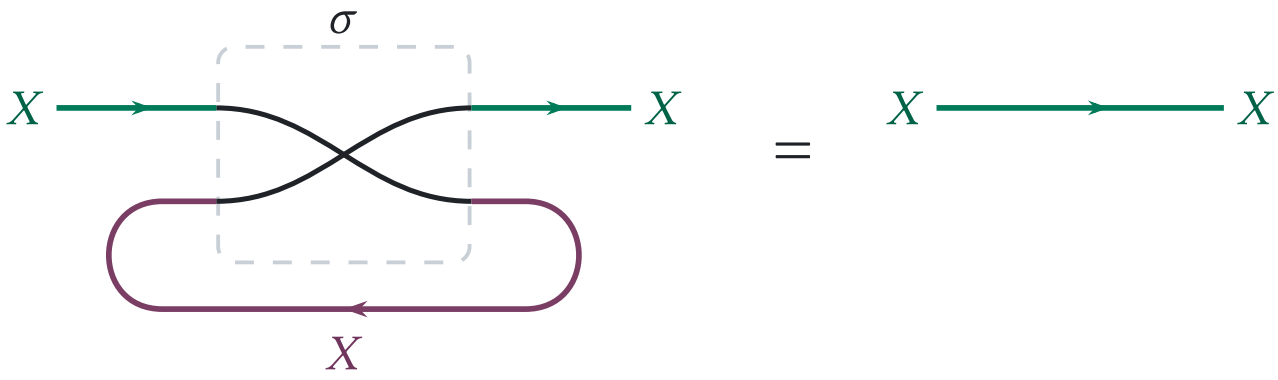
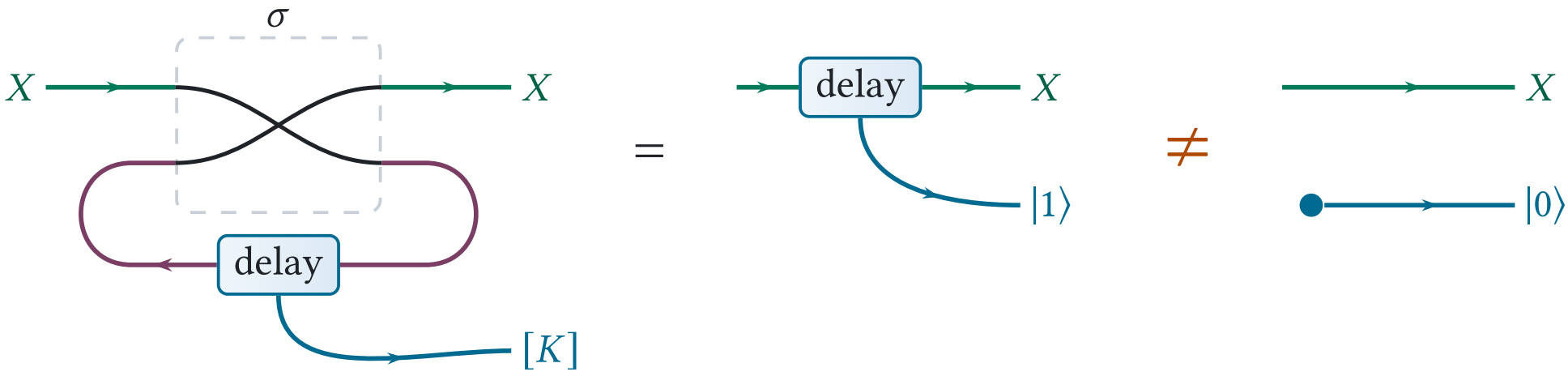
(a) the coherent loop of the swap is the identity



$$\mathrm{Tr}^X(\sigma) = I_X, \text{ yanking}$$

(b) a nonempty input distinguishes the recorded delay



count_K of σ, $K \geq 2$
$\dim X > 0$

$|1\rangle \otimes I$
the identity, one return recorded

$|0\rangle \otimes I$
the identity body

On the swap, the measured loop sends ρ to ρ.